

Physical Chemistry of Materials

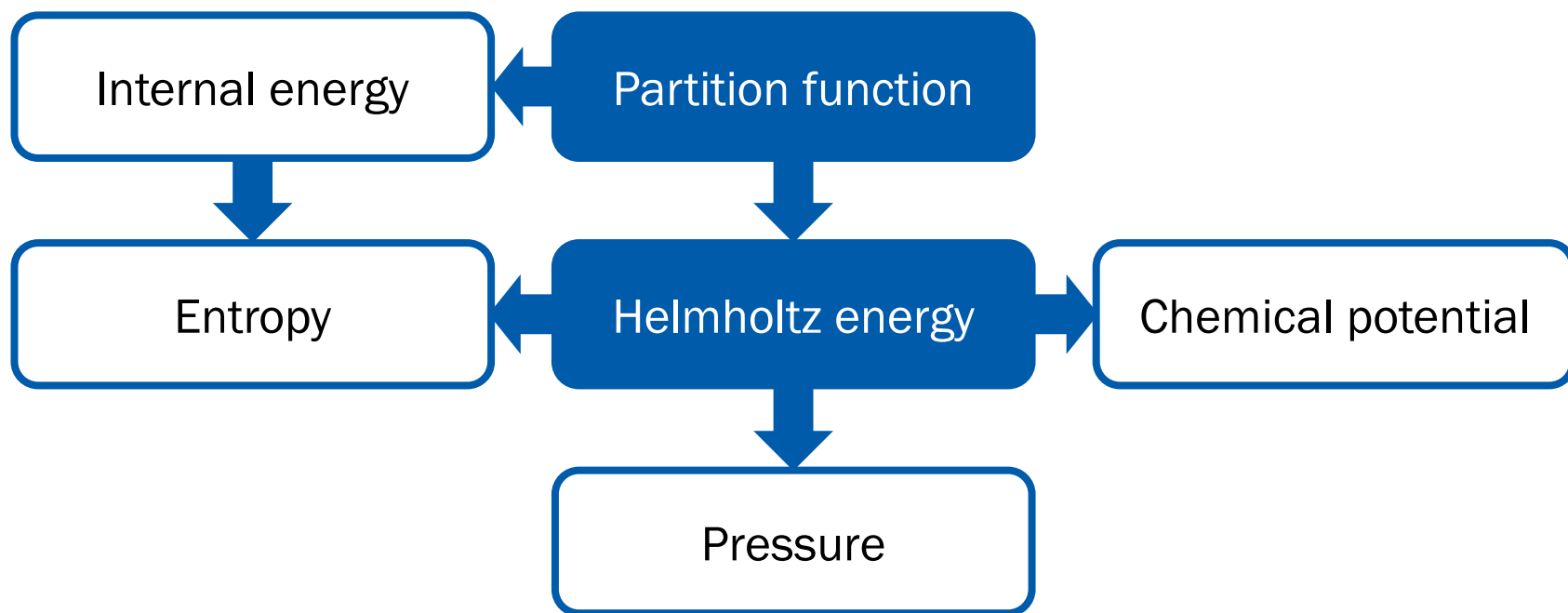
Lecture 06

Ideal Gas

Chemical Equilibrium



Z and Thermodynamics



Multi-Particle PFs

- Independent & distinguishable particles

$$Z = z^N \text{ (} z: \text{ one-particle partition function)}$$

- Independent & indistinguishable particles

$$Z = \frac{1}{N!} z^N$$

- Interacting particles: later

Classical Partition Function

$$z = \sum_s e^{-\beta E(s)} \rightarrow z = \frac{1}{\sigma_0} \int \int e^{-\beta E(x,p)} dx dp$$

for a 1-dimensional single particle

PF for 3D Ideal Gas

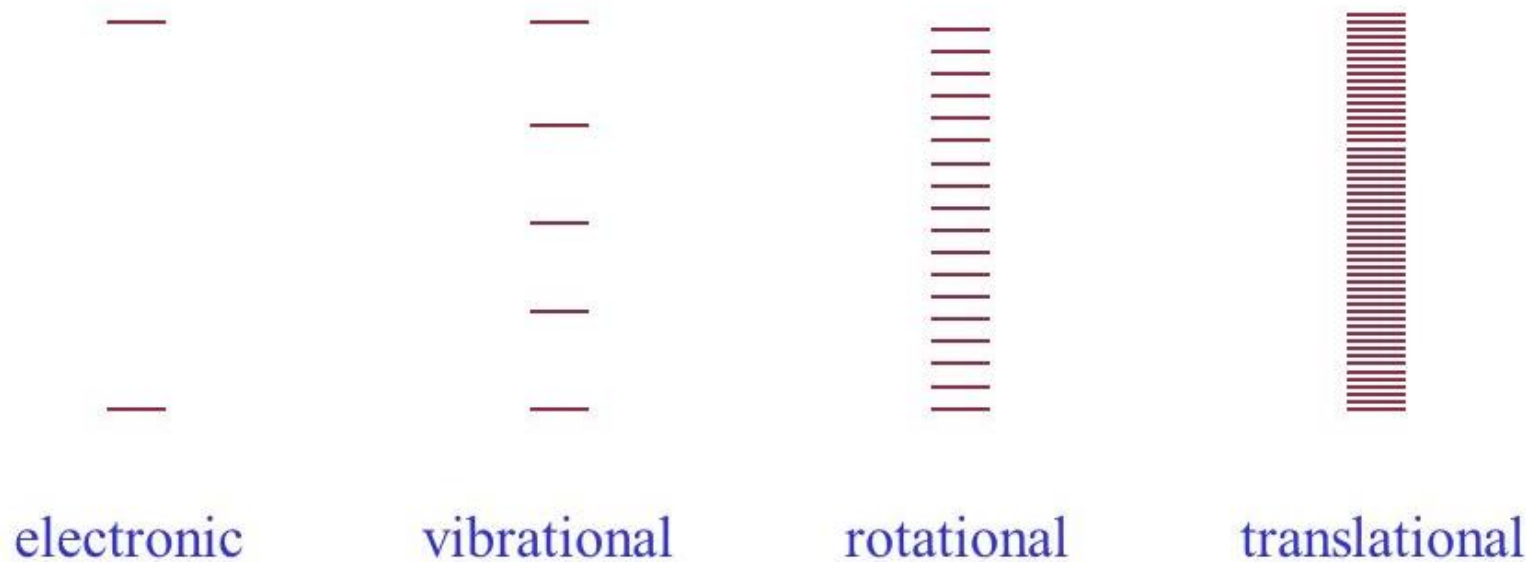
- Classical version

$$Z = \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{\sigma_0^2} \right)^{3N/2}$$

- Quantum version

$$Z = \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2}$$

Molecular Energies in QM



QM Model Systems for Molecules

- **Translation:** particle in a box

$$E_n = \frac{h^2}{8ma^2} n^2$$

- **Vibration:** simple harmonic oscillator

$$E_v = h\nu \left(v + \frac{1}{2} \right)$$

- **Rotation:** rigid rotor

$$E_J = \hbar^2 J(J + 1)/2I \text{ and } g_J = 2J + 1$$

Energy and Partition Function

- Monatomic molecules

$$E = E_{\text{trans}} + E_{\text{elec}}$$

$$Z = Z_{\text{trans}}Z_{\text{elec}}$$

- Diatomic and polyatomic molecules

$$E = E_{\text{trans}} + E_{\text{vib}} + E_{\text{rot}} + E_{\text{elec}}$$

$$Z = Z_{\text{trans}}Z_{\text{vib}}Z_{\text{rot}}Z_{\text{elec}}$$

Electronic Partition Function

At room temperature, almost all molecules are in their **ground electronic states** (no thermal excitation)

$$Z_{\text{elec}} = \sum_{\text{electronic state } s} e^{-E_s/k_B T} \approx g_0 e^{-E_0/k_B T}$$

Translational Partition Function

- Particle in a box
- Already calculated in the previous section

$$z_{\text{trans}} = \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2} V$$

- Consistent for classical and quantum treatments!

Vibrational Partition Function

- Simple harmonic oscillator
- Classical treatment

$$Z_{\text{vib}} = \frac{1}{\sigma_0} \int \int e^{-\beta E(x,p)} dx dp = \frac{1}{h} \int \int e^{-\beta E(x,p)} dx dp$$

$$E(x, p) = \frac{1}{2} k x^2 + \frac{1}{2} m v^2 = \frac{1}{2} k x^2 + \frac{p^2}{2m}$$

$$Z_{\text{vib}} = \frac{1}{h} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\beta \left(\frac{1}{2} k x^2 + \frac{p^2}{2m} \right)} dx dp$$

Classical 1-Dimensional Spring

$$Z_{\text{vib}} = \frac{1}{h} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\beta \left(\frac{1}{2} k x^2 + \frac{p^2}{2m} \right)} dx dp$$

$$Z_{\text{vib}} = \frac{1}{h} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{\beta k x^2}{2} - \frac{\beta p^2}{2m}} dx dp$$

$$Z_{\text{vib}} = \frac{1}{h} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{\beta k x^2}{2}} \times e^{-\frac{\beta p^2}{2m}} dx dp$$

$$Z_{\text{vib}} = \frac{1}{h} \left[\int_{-\infty}^{\infty} e^{-\frac{\beta k x^2}{2}} dx \right] \left[\int_{-\infty}^{\infty} e^{-\frac{\beta p^2}{2m}} dp \right]$$

Gaussian Integrals

$$Z_{\text{vib}} = \frac{1}{h} \left[\int_{-\infty}^{\infty} e^{-\frac{\beta k x^2}{2}} dx \right] \left[\int_{-\infty}^{\infty} e^{-\frac{\beta p^2}{2m}} dp \right]$$

Using $\int_{-\infty}^{\infty} e^{-\alpha x^2} dx = \sqrt{\pi/\alpha},$

$$Z_{\text{vib}} = \frac{1}{h} \sqrt{\frac{2\pi}{\beta k}} \sqrt{\frac{2m\pi}{\beta}} = \frac{2\pi}{\beta h} \sqrt{\frac{m}{k}}$$

Vibrational Partition Function

- Simple harmonic oscillator
- Classical treatment

$$z_{\text{vib}} = \frac{2\pi}{\beta h} \sqrt{\frac{m}{k}} = \frac{k_{\text{B}}T}{h\nu}$$

- Quantum treatment

$$z_{\text{vib}} = \sum_{v=0}^{\infty} e^{-\beta E_v} = \sum_{v=0}^{\infty} e^{-\beta h\nu \left(v + \frac{1}{2}\right)}$$

Quantum Version of z_{vib}

$$z_{\text{vib}} = \sum_{v=0}^{\infty} e^{-\beta h \nu \left(v + \frac{1}{2}\right)}$$

Let $x = e^{-\beta h \nu}$; for $x < 1$,

$$z_{\text{vib}} = x^{1/2} \sum_{v=0}^{\infty} x^v = \frac{x^{1/2}}{1 - x}$$

$$\therefore z_{\text{vib}} = \frac{e^{-\beta h \nu / 2}}{1 - e^{-\beta h \nu}}$$

Comparison of Two Versions

- Classical treatment

$$Z_{\text{vib,CM}} = \frac{k_B T}{h\nu}$$

- Quantum treatment

$$Z_{\text{vib,QM}} = \frac{e^{-h\nu/2k_B T}}{1 - e^{-h\nu/k_B T}}$$

- For $T \rightarrow \infty$ (classical limit), using $e^{-x} \approx 1 - x$ for $x \ll 1$,

$$Z_{\text{vib,QM}} \approx \frac{k_B T}{h\nu}$$

Rotational Partition Function

- Rigid rotor
- Classical treatment (requires messy math)

$$z_{\text{rot}} = \frac{8\pi^2 I}{\beta h^2}$$

- Quantum treatment

$$z_{\text{rot}} = \sum_{J=0}^{\infty} g_J e^{-\beta E_J} = \sum_{J=0}^{\infty} (2J + 1) e^{-\beta \hbar^2 J(J+1)/2I}$$

Quantum Version of z_{rot}

$$z_{\text{rot}} = \sum_{J=0}^{\infty} (2J + 1) e^{-\beta \hbar^2 J(J+1)/2I}$$

Take a continuum approximation (at high T):

$$z_{\text{rot}} = \int_0^{\infty} (2J + 1) e^{-\beta \hbar^2 J(J+1)/2I} dJ$$

Letting $x = J(J + 1)$,

$$z_{\text{rot}} = \int_0^{\infty} e^{-\beta \hbar^2 x/2I} dx = \frac{2I}{\beta \hbar^2}$$

Symmetry Number

- Homonuclear diatomic molecule: 180° rotation leads to the same configuration

→ Correction required!

- Symmetry number σ

$$Z_{\text{rot}} = \frac{2Ik_{\text{B}}T}{\sigma\hbar^2}$$

“Number of indistinguishable orientations of the molecule”

Linear Polyatomic Molecules

- Energy

$$E = E_{\text{trans}} + E_{\text{rot}} + \sum_j E_{\text{vib},j} + E_{\text{elec}}$$

- Partition functions

$$z = \underbrace{\left(\frac{2\pi M k_B T}{h^2}\right)^{3/2} V}_{z_{\text{trans}}} \cdot \underbrace{\frac{2Ik_B T}{\sigma \hbar^2}}_{z_{\text{rot}}} \cdot \underbrace{\prod_j \frac{e^{-h\nu_j/2k_B T}}{1 - e^{-h\nu_j/k_B T}}}_{z_{\text{vib}}} \cdot \underbrace{g_0 e^{-E_0/k_B T}}_{z_{\text{elec}}}$$

$$Z = \frac{1}{N!} z^N = \frac{1}{N!} (z_{\text{trans}} z_{\text{rot}} z_{\text{vib}} z_{\text{elec}})^N$$

Linear Polyatomic Molecules

$$Z = \frac{1}{N!} (z_{\text{trans}} z_{\text{rot}} z_{\text{vib}} z_{\text{elec}})^N$$

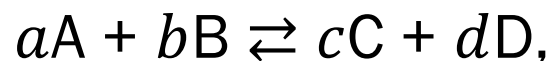
- Internal energy

$$U = - \frac{\partial \ln Z}{\partial \beta}$$

$$U = \frac{5}{2} N k_B T + \sum_j \left(\frac{N h \nu_j}{2} + \frac{N h \nu_j}{e^{h \nu_j / k_B T} - 1} \right) + N E_0$$

Law of Mass Action

For a chemical equilibrium

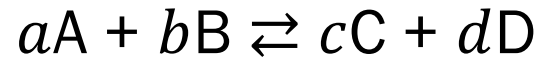


the **equilibrium constant** is constant at fixed temperature

$$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

Why?

Extent of Reaction



Define ξ such that

$$dn_A = -a d\xi$$

$$dn_B = -b d\xi$$

$$dn_C = +c d\xi$$

$$dn_D = +d d\xi$$

Helmholtz Energy

$$dA = -SdT - pdV + \sum_i \mu_i dn_i$$

For fixed temperature and volume,

$$dA = \sum_i \mu_i dn_i = (-a\mu_A - b\mu_B + c\mu_C + d\mu_D)d\xi$$

At equilibrium, $dA = 0$ and

$$-a\mu_A - b\mu_B + c\mu_C + d\mu_D = 0$$

Chemical Potential

$$\mu = \left(\frac{\partial A}{\partial n} \right)_{T,V} = -k_B T \left(\frac{\partial \ln Z}{\partial n} \right)_{T,V} = -RT \left(\frac{\partial \ln Z}{\partial N} \right)_{T,V}$$

For N molecules, $Z = \frac{1}{N!} z^N$ and

$$\ln Z \approx N \ln z - N \ln N + N$$

Hence,

$$\mu = -RT \ln z + RT \ln N + RT - RT = -RT \ln \left(\frac{z}{N} \right)$$

Partition Functions in Action

$$-a\mu_A - b\mu_B + c\mu_C + d\mu_D = 0$$

$$\mu_i = -RT \ln \left(\frac{z_i}{N_i} \right)$$

Substitution leads to

$$RT \left[a \ln \left(\frac{z_A}{N_A} \right) + b \ln \left(\frac{z_B}{N_B} \right) - c \ln \left(\frac{z_C}{N_C} \right) - d \ln \left(\frac{z_D}{N_D} \right) \right] = 0$$

$$\therefore \frac{N_C^c N_D^d}{N_A^a N_B^b} = \frac{z_C^c z_D^d}{z_A^a z_B^b}$$

Equilibrium Constant

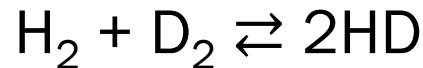
$$\frac{N_C^c N_D^d}{N_A^a N_B^b} = \frac{z_C^c z_D^d}{z_A^a z_B^b}$$

$$\frac{(N_C/V)^c (N_D/V)^d}{(N_A/V)^a (N_B/V)^b} = \frac{(z_C/V)^c (z_D/V)^d}{(z_A/V)^a (z_B/V)^b}$$

The left-hand side is the equilibrium constant:

$$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b} = \frac{(z_C/V)^c (z_D/V)^d}{(z_A/V)^a (z_B/V)^b}$$

K from Partition Function



$$K_c = \frac{[\text{HD}]^2}{[\text{H}_2][\text{D}_2]} = \frac{(z_{\text{HD}}/V)^2}{(z_{\text{H}_2}/V)(z_{\text{D}_2}/V)} = \frac{z_{\text{HD}}^2}{z_{\text{H}_2}z_{\text{D}_2}}$$

For a diatomic molecule, recall that

$$Z = Z_{\text{trans}}Z_{\text{rot}}Z_{\text{vib}}Z_{\text{elec}}$$

$$= \left(\frac{2\pi M k_B T}{h^2} \right)^{3/2} V \cdot \frac{2Ik_B T}{\sigma \hbar^2} \cdot \frac{e^{-h\nu/2k_B T}}{1 - e^{-h\nu/k_B T}} \cdot g_0 e^{-E_0/k_B T}$$

≈ 1 (at RT)

Equilibrium Constant

Substitution leads to

$$K_c = \left(\frac{M_{\text{HD}}^2}{M_{\text{H}_2} M_{\text{D}_2}} \right)^{3/2} \frac{I_{\text{HD}}^2}{I_{\text{H}_2} I_{\text{D}_2}} \frac{\sigma_{\text{H}_2} \sigma_{\text{D}_2}}{\sigma_{\text{HD}}^2} e^{-\beta h(2\nu_{\text{HD}} - \nu_{\text{H}_2} - \nu_{\text{D}_2})/2}$$

Using $M_{\text{AB}} = m_{\text{A}} + m_{\text{B}}$ and $I_{\text{AB}} = \frac{m_{\text{A}} m_{\text{B}}}{m_{\text{A}} + m_{\text{B}}} R_{\text{AB}}^2$,

$$K_c = \left(\frac{9}{8} \right)^{3/2} \times \frac{8}{9} \times 4 \times e^{-\beta h(2\nu_{\text{HD}} - \nu_{\text{H}_2} - \nu_{\text{D}_2})/2}$$

$$\therefore K_c = 4.24 \times e^{-(77.7 \text{ K})/T}$$

Theory vs Experiment

T (K)	K_c (theory)	K_c (exp)
195	2.84	2.92
273	3.18	3.24
298	3.26	3.28
383	3.46	3.50
543	3.67	3.85